

CBSE Class 12 Chemistry — Half-Yearly Predicted Paper (Set 9 of 10)

50 Most-Probable PYQs & Board-Pattern Practice Questions

Section A: 1-Mark Questions (MCQs / Assertion-Reason)

- Q1.** The value of Henry's constant (K_H) is: (a) Greater for gases with higher solubility (b) Greater for gases with lower solubility (c) Constant for all gases at all temperatures (d) Unrelated to the solubility of the gas [Solutions] [PYQ 2020] [Confidence: Very High]
- Q2.** Which of the following halogens is typically exchanged in the Swarts reaction to form alkyl fluorides? (a) Chlorine or Bromine (b) Iodine (c) Astatine (d) Only Chlorine [Haloalkanes and Haloarenes] [Practice] [Confidence: High]
- Q3.** In a zero-order reaction, the plot of concentration of reactant $[R]$ versus time (t) yields a straight line. What does the y-intercept of this line represent? (a) The rate constant (k) (b) The half-life ($t_{1/2}$) (c) The initial concentration $[R]_0$ (d) Zero [Chemical Kinetics] [PYQ 2023] [Confidence: Very High]
- Q4.** Which of the following transition metal ions is colourless in an aqueous solution? (a) Ti^{3+} (b) V^{3+} (c) Cu^{2+} (d) Sc^{3+} [The d- and f-Block Elements] [PYQ 2017] [Confidence: Very High]
- Q5.** The denticity of the oxalate ligand ($C_2O_4^{2-}$) in a coordination complex is: (a) Unidentate (b) Bidentate (c) Tridentate (d) Hexadentate [Coordination Compounds] [Practice] [Confidence: Very High]
- Q6.** The conversion of an alkyl halide to an ether by reacting it with sodium alkoxide is known as: (a) Wurtz reaction (b) Finkelstein reaction (c) Williamson's continuous etherification process (d) Williamson synthesis [Alcohols, Phenols and Ethers] [PYQ 2015] [Confidence: Very High]
- Q7.** Which of the following alcohols yields a ketone upon oxidation with Pyridinium Chlorochromate (PCC)? (a) Ethanol (b) Propan-1-ol (c) Propan-2-ol (d) 2-Methylpropan-2-ol [Alcohols, Phenols and Ethers] [Practice] [Confidence: High]
- Q8.** The specific reagent mixture used in the Clemmensen reduction to convert a ketone to an alkane is: (a) $SnCl_2 + HCl$ (b) $Zn(Hg) + conc. HCl$ (c) $NH_2NH_2 + KOH$ (d) $H_2/Pd - BaSO_4$ [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2016] [Confidence: Very High]
- Q9.** The standard electrode potential of a Standard Hydrogen Electrode (SHE) is assigned a value of: (a) $1.0 V$ (b) $0.0 V$ (c) $-1.0 V$ (d) $0.5 V$ [Electrochemistry] [Practice] [Confidence: High]
- Q10.** The unit of the cell constant (G^*) of a conductivity cell is: (a) cm (b) cm^{-1} (c) $S cm^{-1}$ (d) Ωcm [Electrochemistry] [PYQ 2014] [Confidence: Very High]
- Q11.** For a reaction with rate law $Rate = k[A]^2[B]^0$, what happens to the rate if the concentration of B is doubled? (a) Rate is doubled (b) Rate is quadrupled (c) Rate remains unchanged (d) Rate is halved [Chemical Kinetics] [Practice] [Confidence: Very High]
- Q12.** The reaction of toluene with Cl_2 in the presence of $FeCl_3$ in dark gives: (a) Benzyl chloride (b) m-Chlorotoluene (c) o- and p-Chlorotoluene (d) Trichloromethylbenzene [Haloalkanes and Haloarenes] [PYQ Term-1 2021] [Confidence: Very High]

Directions for Q13 to Q16: A statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

- Q13.** Assertion (A): Zinc ($Z = 30$), Cadmium ($Z = 48$), and Mercury ($Z = 80$) are generally not considered true transition elements. Reason (R): They possess completely filled d-orbitals in their ground state as well as in their common oxidation states. [The d- and f-Block Elements] [PYQ 2014] [Confidence: Very High]
- Q14.** Assertion (A): Ethers possess significantly lower boiling points compared to their isomeric alcohols. Reason (R): Ethers lack an oxygen-bound hydrogen atom and thus cannot form intermolecular hydrogen bonds. [Alcohols, Phenols and Ethers] [PYQ 2015] [Confidence: Very High]
- Q15.** Assertion (A): Benzaldehyde readily undergoes the Aldol condensation reaction when treated with dilute alkali. Reason (R): Benzaldehyde completely lacks an alpha-hydrogen atom on the carbon adjacent to the carbonyl group. [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2023] [Confidence: Very High]

Q16. Assertion (A): The coordination complex $[\text{Fe}(\text{CN})_6]^{4-}$ is diamagnetic in nature. Reason (R): Cyanide (CN^-) is a strong field ligand and forces the complete pairing of 3d electrons in the Fe^{2+} ion. [Coordination Compounds] [PYQ 2018] [Confidence: Very High]

Section B: 2-Mark Questions (Short Answer)

Q17. Define the term *Molal Elevation Constant* (K_b). State its SI unit. [Solutions] [PYQ 2014] [Confidence: Very High]

Q18. Explain the following observation in electrochemistry: The conductivity (κ) of an electrolytic solution decreases with dilution, whereas its molar conductivity (Λ_m) increases with dilution. [Electrochemistry] [PYQ Term-2 2022] [Confidence: Very High]

Q19. A first-order reaction takes 10 minutes for 20% completion. Calculate the specific rate constant (k) for this reaction. (Given: $\log 10 = 1.0000$, $\log 8 = 0.9031$). [Chemical Kinetics] [Practice] [Confidence: High]

Q20. Account for the following strict observations in transition metal chemistry: (a) The enthalpies of atomization of transition metals are remarkably high. (b) Transition metals readily form stable interstitial compounds. [The d- and f-Block Elements] [PYQ 2018] [Confidence: Very High]

Q21. Using IUPAC nomenclature norms, write the exact formulae for the following complexes: (a) Diamminechloridonitrito-N-platinum(II) (b) Potassium trioxalatoferrate(III) [Coordination Compounds] [PYQ 2017] [Confidence: High]

Section C: 3-Mark Questions (Short Answer)

Q22. Calculate the boiling point of an aqueous solution prepared by dissolving 18 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in 100 g of water at 1.013 bar pressure. (Given: K_b for water = $0.52 \text{ K kg mol}^{-1}$, Molar mass of glucose = 180 g mol^{-1} , Boiling point of pure water = 373.15 K). [Solutions] [Practice] [Confidence: Very High]

Q23. Calculate the standard Gibbs free energy change (ΔG°) and the equilibrium constant ($\log K_c$) for the following reaction at 298 K: $\text{Cu}(s) + 2\text{Ag}^+(aq) \rightarrow \text{Cu}^{2+}(aq) + 2\text{Ag}(s)$ (Given: $E^\circ_{\text{cell}} = 0.46 \text{ V}$, $1F = 96500 \text{ C mol}^{-1}$, $2.303RT/F = 0.0591 \text{ V}$). [Electrochemistry] [PYQ 2015] [Confidence: Very High]

Q24. Show mathematically that for a strictly first-order reaction, the time required for 75% completion is exactly twice the time required for 50% completion (its half-life). [Chemical Kinetics] [PYQ 2016] [Confidence: Very High]

Q25. Give distinct chemical reasons to account for the following observations: (a) The dipole moment of chlorobenzene is notably lower than that of cyclohexyl chloride. (b) $\text{S}_{\text{N}}1$ reactions are strongly preferred by tertiary alkyl halides. (c) A true racemic mixture is optically inactive. [Haloalkanes and Haloarenes] [PYQ 2017 / 2016] [Confidence: Very High]

Q26. Write the exact balanced chemical equations for the laboratory preparation of phenol from cumene (isopropylbenzene). [Alcohols, Phenols and Ethers] [PYQ 2020] [Confidence: High]

Q27. Give simple, observable chemical tests to distinctly differentiate between the following pairs of compounds: (a) Propanal and Propanone (b) Phenol and Benzoic acid (c) Acetophenone and Benzophenone [Aldehydes, Ketones and Carboxylic Acids / Alcohols] [PYQ 2014 / Practice] [Confidence: Very High]

Q28. Write the balanced chemical equations to clearly illustrate the following named organic reactions: (a) Esterification reaction (b) Hell-Volhard-Zelinsky (HVZ) reaction (c) Decarboxylation of a carboxylic acid [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2015 / 2018] [Confidence: Very High]

Section D: 4-Mark Questions (Case-Study)

Q29. Case Study: Catalytic and Redox Properties of Transition Metals The d-block elements (transition metals) and their compounds are extensively utilized as catalysts in numerous industrial and biological processes. This catalytic capability is primarily attributed to their ability to exhibit variable oxidation states and form unstable intermediate complexes with reactants. Furthermore, transition metals display a wide variety of redox behaviours. The stability of a particular oxidation state heavily depends on the electronic configuration of the resulting ion, particularly the presence of exactly half-filled or completely filled d-subshells. (a) Chemically, why do transition metals act as excellent catalysts? (1 Mark) (b) Identify the transition element in the 3d series that exhibits the maximum number of oxidation states. What is its highest oxidation state? (1 Mark) (c) Account for the following: Although both Cr^{2+} and Mn^{3+} ions have the exact same

d^4 electronic configuration, Cr^{2+} is a strong reducing agent whereas Mn^{3+} is a strong oxidizing agent. (2 Marks) [The d- and f-Block Elements] [CBSE SQP 2022 Themes] [Confidence: Very High]

Q30. Case Study: Cleavage of Ethers by Halogen Acids Ethers are relatively inert organic compounds and are generally unreactive towards bases, oxidizing agents, and reducing agents under normal conditions. However, they undergo cleavage of the $\text{C} - \text{O}$ bond when heated with an excess of strong halogen acids like HI or HBr . The cleavage of alkyl aryl ethers (like anisole) uniquely results in the formation of a phenol and an alkyl halide, rather than an aryl halide and an alcohol, due to the high stability of the aryl-oxygen bond. (a) Write the exact chemical structures of the products formed when anisole (methoxybenzene) is heated with an excess of HI . (1 Mark) (b) Provide a structural reason why the cleavage of anisole exclusively gives phenol and methyl iodide, and never iodobenzene and methanol. (1 Mark) (c) Predict the major organic products when tert-butyl methyl ether reacts with one equivalent of hot HI . State the specific mechanism ($\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$) followed in this case. (2 Marks) [Alcohols, Phenols and Ethers] [PYQ 2015 / 2018 Themes] [Confidence: High]

Section E: 5-Mark Questions (Long Answer)

Q31. (a) State Kohlrausch's law of independent migration of ions. (1 Mark) (b) The limiting molar conductivities (Λ_m°) for HCl , NaCl , and CH_3COONa are **425.9**, **126.4**, and **91.0 $\text{S cm}^2 \text{mol}^{-1}$** respectively at **298 K**. Calculate the limiting molar conductivity of acetic acid (CH_3COOH). (2 Marks) (c) Define the term *Osmotic Pressure*. Why is the measurement of osmotic pressure preferred over other colligative properties for determining the molar masses of macromolecules like proteins? (2 Marks) [Electrochemistry / Solutions] [PYQ 2015 / 2020] [Confidence: Very High]

Q32. (a) Using Crystal Field Theory (CFT), write the electronic configuration of a d^4 central metal ion if the crystal field splitting energy is strictly less than the pairing energy ($\Delta_o < P$). (1 Mark) (b) Using Valence Bond Theory (VBT), deduce the state of hybridization, the geometry, and the magnetic property of the complex ion $[\text{CoF}_6]^{3-}$. (Given: Atomic number of Co = 27). (2 Marks) (c) Write the complete chemical equations to carry out the following conversions logically: (i) Propene to 1-Bromopropane (ii) Chlorobenzene to Biphenyl (Fittig reaction) (2 Marks) [Coordination Compounds / Haloalkanes] [PYQ 2016 / 2019] [Confidence: Very High]

Q33. An organic compound **A** having the molecular formula $\text{C}_3\text{H}_6\text{O}$ is highly resistant to mild oxidation but undergoes catalytic reduction to form a secondary alcohol **B** ($\text{C}_3\text{H}_8\text{O}$). Compound **B** when heated with concentrated sulphuric acid at **443 K** undergoes dehydration to form a gaseous alkene **C** (C_3H_6). Alkene **C** reacts with HBr according to Markovnikov's rule to yield an alkyl bromide **D**. (a) Deduce the exact structures and standard IUPAC names of compounds **A**, **B**, **C**, and **D**. (4 Marks) (b) Write the chemical equation for the reduction of compound **A** to compound **B**. (1 Mark) [Aldehydes, Ketones and Carboxylic Acids / Alcohols] [PYQ 2016 / Practice] [Confidence: High]

Answer Key

Q1. (b) Greater for gases with lower solubility. ($P = K_H \cdot x$. Higher K_H means lower mole fraction x in solution for a given pressure). **Q2.** (a) Chlorine or Bromine. (Alkyl chlorides/bromides are reacted with metallic fluorides like AgF). **Q3.** (c) The initial concentration $[\text{R}]_0$. (Equation: $[\text{R}] = -kt + [\text{R}]_0$. It matches $y = mx + c$, so intercept $c = [\text{R}]_0$). **Q4.** (d) Sc^{3+} . (It has a $3d^0$ configuration with no unpaired electrons to undergo d-d transitions). **Q5.** (b) Bidentate. (It donates two pairs of electrons through its two oxygen atoms). **Q6.** (d) Williamson synthesis. **Q7.** (c) Propan-2-ol. (A secondary alcohol oxidizes to a ketone). **Q8.** (b) $\text{Zn(Hg)} + \text{conc. HCl}$. **Q9.** (b) **0.0 V**. **Q10.** (b) cm^{-1} (or m^{-1}). **Q11.** (c) Rate remains unchanged. (Since the order with respect to B is 0, changing its concentration does not affect the rate). **Q12.** (c) o- and p-Chlorotoluene. (Electrophilic aromatic substitution). **Q13.** (a) Both A and R are true and R is the correct explanation of A. **Q14.** (a) Both A and R are true and R is the correct explanation of A. **Q15.** (d) A is false but R is true. (Benzaldehyde does *not* undergo Aldol condensation precisely because it lacks an alpha-hydrogen; it undergoes the Cannizzaro reaction). **Q16.** (a) Both A and R are true and R is the correct explanation of A.

Q17. Molal elevation constant (K_b) is the elevation in boiling point observed when one mole of a non-volatile solute is dissolved in **1 kg** of the solvent. Its SI unit is **K kg mol^{-1}** . **Q18.** Conductivity (κ) decreases with dilution because the total number of ions per unit volume carrying the current decreases. Molar conductivity (Λ_m) increases with dilution because the total volume **V** of the solution containing one mole of electrolyte increases drastically, outweighing the decrease in κ . **Q19.** 20% complete means 80% remains ($[\text{R}] = 0.8[\text{R}]_0$).

$$k = \frac{2.303}{t} \log \left(\frac{[\text{R}]_0}{[\text{R}]} \right) = \frac{2.303}{10} \log \left(\frac{100}{80} \right) = 0.2303 \times (\log 10 - \log 8).$$

$$k = 0.2303 \times (1.0000 - 0.9031) = 0.2303 \times 0.0969 = 0.0223 \text{ min}^{-1}. \text{ Q20. (a) High enthalpies of atomization are}$$

due to the presence of a large number of unpaired d-electrons, leading to extremely strong interatomic metallic bonding. (b) Transition metals form interstitial compounds because small atoms like H, C, and N easily get trapped in the empty spaces (interstices) of their crystal lattices. **Q21.** (a) $[Pt(NH_3)_2Cl(ONO)]$ (b) $K_3[Fe(C_2O_4)_3]$.

Q22. Moles of glucose = $18/180 = 0.1 \text{ mol}$. Mass of solvent (water) = $100 \text{ g} = 0.1 \text{ kg}$. Molality (m) = $0.1/0.1 = 1.0 \text{ m}$. $\Delta T_b = K_b \times m = 0.52 \times 1.0 = 0.52 \text{ K}$. Boiling point of solution = $T_b^\circ + \Delta T_b = 373.15 + 0.52 = 373.67 \text{ K}$. **Q23.** $n = 2$. $\Delta G^\circ = -nFE_{cell}^\circ = -2 \times 96500 \times 0.46 = -88,780 \text{ J mol}^{-1}$ or $-88.78 \text{ kJ mol}^{-1}$.

$E_{cell}^\circ = \frac{0.0591}{n} \log K_c \Rightarrow 0.46 = \frac{0.0591}{2} \log K_c \Rightarrow \log K_c = \frac{0.92}{0.0591} \approx 15.56$. **Q24.** For a 1st order reaction:

$$t = \frac{2.303}{k} \log \left(\frac{100}{100-x} \right). t_{50\%} = \frac{2.303}{k} \log \left(\frac{100}{50} \right) = \frac{2.303}{k} \log 2.$$

$t_{75\%} = \frac{2.303}{k} \log \left(\frac{100}{25} \right) = \frac{2.303}{k} \log 4 = \frac{2.303}{k} \log(2^2) = 2 \times \left[\frac{2.303}{k} \log 2 \right]$. Therefore, $t_{75\%} = 2 \times t_{50\%}$. **Q25.** (a) In chlorobenzene, the sp^2 hybridized carbon is more electronegative than the sp^3 carbon in cyclohexyl chloride, decreasing the polarity. Also, +R resonance effect of Cl opposes the -I effect. (b) Tertiary alkyl halides form stable 3° carbocations, facilitating the S_N1 pathway. (c) A racemic mixture contains equal amounts of dextro and laevo enantiomers, so their optical rotations perfectly cancel each other out (external compensation). **Q26.** Step 1: Cumene + $O_2 \rightarrow$ Cumene hydroperoxide. Step 2: Cumene hydroperoxide $\xrightarrow{H^+, H_2O}$ Phenol + Acetone (CH_3COCH_3). **Q27.** (a) Add Tollens' reagent. Propanal (aldehyde) forms a silver mirror; Propanone (ketone) does not. (b) Add neutral $FeCl_3$. Phenol gives a violet colouration; Benzoic acid gives a buff-coloured precipitate. (c) Iodoform test. Acetophenone gives a yellow ppt of CHI_3 with $I_2/NaOH$; Benzophenone does not. **Q28.** (a) $CH_3COOH + C_2H_5OH \rightleftharpoons (inH^+)CH_3COOC_2H_5 + H_2O$. (b)

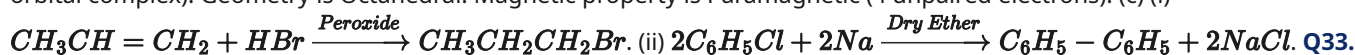
$CH_3COOH + Cl_2 \xrightarrow{Red P, H_2O} Cl-CH_2-COOH + HCl$. (c) $CH_3COONa + NaOH \xrightarrow{CaO, \Delta} CH_4 \uparrow + Na_2CO_3$.

Q29. (a) They provide a large surface area for adsorption and can easily adopt multiple oxidation states to form intermediate complexes with lower activation energies. (b) Manganese (Mn). Highest state is +7 (e.g., in $KMnO_4$). (c) Cr^{2+} oxidizes to Cr^{3+} , gaining an extremely stable exactly half-filled t_{2g}^3 orbital configuration in aqueous solutions. Mn^{3+} reduces to Mn^{2+} , gaining a highly stable spherically symmetrical half-filled $3d^5$ configuration. **Q30.** (a) Phenol (C_6H_5OH) and Methyl iodide (CH_3I). (b) The $O-C_6H_5$ (aryl-oxygen) bond acquires partial double bond character due to resonance and is much stronger than the $O-CH_3$ (alkyl-oxygen) bond. Thus, nucleophilic attack by I^- strictly breaks the $O-CH_3$ bond. (c) Products: Methanol (CH_3OH) and tert-butyl iodide ($(CH_3)_3C-I$). It follows the S_N1 mechanism because the tert-butyl carbocation formed intermediate is highly stable.

Q31. (a) The limiting molar conductivity of an electrolyte is equal to the sum of the individual contributions of the limiting molar conductivities of its constituent cations and anions. (b)

$$\Lambda_m^\circ(CH_3COOH) = \Lambda_m^\circ(CH_3COONa) + \Lambda_m^\circ(HCl) - \Lambda_m^\circ(NaCl) = 91.0 + 425.9 - 126.4 = 390.5 \text{ S cm}^2 \text{ mol}^{-1}$$

. (c) Osmotic pressure is the extra pressure that must be applied to the solution side to prevent the flow of solvent molecules into it through a semi-permeable membrane. It is preferred for macromolecules because they are unstable at high temperatures and have poor solubility; osmotic pressure gives a large, easily measurable value at room temperature ($\pi = CRT$). **Q32.** (a) $t_{2g}^3 e_g^1$. (Since $\Delta_o < P$, it is a weak field case and electrons remain unpaired). (b) Co is $3d^7 4s^2$. Co^{3+} is $3d^6$. F^- is a weak field ligand, so pairing does not occur. Outer 4d orbitals are used. Hybridization is $sp^3 d^2$ (outer orbital complex). Geometry is Octahedral. Magnetic property is Paramagnetic (4 unpaired electrons). (c) (i)



(a) **A** is Propanone / Acetone (CH_3COCH_3). **B** is Propan-2-ol ($CH_3-CH(OH)-CH_3$). **C** is Propene ($CH_3-CH=CH_2$). **D** is 2-Bromopropane ($CH_3-CH(Br)-CH_3$). (b) Reduction:

